The $(3, 5)$ Odd-Walk Inequality
Entropy on Finite Graphs, and Relative Entropy on Graphons

Abstract

Section 1 is the Blekherman–Raymond entropy proof of $t(P_5, W)^3 \geq t(P_3, W)^5$ for finite host graphs. Section 2 transcribes the same argument — the same three folds of P_5 onto P_3, the same edge multiplicities $(5, 5, 5)$ and internal-vertex multiplicities $(1, 5, 5, 1)$ — directly into the graphon setting, with Shannon entropy against counting measure replaced by relative entropy against μ. The transcription needs no finite host, no sampling and no approximation step, and it is valid on an arbitrary probability space. That is what the catalogue requires, and it is what the Lean formalization follows.

1 The finite case

Proposition 1.1 (Blekherman–Raymond, P_3–P_5 case)**.** *Let P_k denote the path with k edges. Then, for every graphon W,*

$$t(P_5, W)^3 \geq t(P_3, W)^5.$$

Proof. We first prove the inequality for a finite graph G on n vertices. Write

$$\hom(P_k, G) := |\operatorname{Hom}(P_k, G)|.$$

The desired inequality

$$t(P_5, G)^3 \geq t(P_3, G)^5$$

is equivalent to

$$n^2 \hom(P_5, G)^3 \geq \hom(P_3, G)^5,$$

since P_5 has 6 vertices and P_3 has 4 vertices.

If G has no edges, then both sides vanish, so there is nothing to prove. Assume henceforth that G has at least one edge.

Let

$$X = (X_0, X_1, X_2, X_3)$$

be a uniformly random homomorphism from P_3 to G. Thus X is a uniformly random walk of length 3 in G, and therefore

$$H(X_0, X_1, X_2, X_3) = \log |\operatorname{Hom}(P_3, G)|.$$

The base of the logarithm is irrelevant.

We shall construct a random homomorphism from

$$F := P_0 \sqcup P_0 \sqcup P_5 \sqcup P_5 \sqcup P_5$$

to G whose entropy is exactly $5H(X)$. Since

$$|\operatorname{Hom}(F, G)| = n^2 \hom(P_5, G)^3,$$

the entropy bound

$$H(Y) \le \log |\operatorname{Hom}(F, G)|$$

will then give the result.

Step 1: the Markov property of the uniform random walk.

The law of X is uniform on all quadruples (x_0, x_1, x_2, x_3) satisfying

$$x_0x_1,\ x_1x_2,\ x_2x_3 \in E(G).$$

Consequently, conditioned on X_i, the variables on the two sides of X_i are conditionally independent. In particular,

$$X_0 \perp (X_2, X_3) \mid X_1, \qquad (X_0, X_1) \perp X_3 \mid X_2.$$

Hence the entropy of X satisfies the tree entropy identity

$$\begin{aligned} H(X) = H(X_0, X_1) + H(X_1, X_2) + H(X_2, X_3) \\ - H(X_1) - H(X_2). \end{aligned} \tag{1}$$

Indeed, by the chain rule and the conditional independences,

$$\begin{aligned} H(X_0, X_1, X_2, X_3) &= H(X_1) + H(X_0 \mid X_1) + H(X_2 \mid X_1) + H(X_3 \mid X_2) \\ &= H(X_0, X_1) + H(X_1, X_2) + H(X_2, X_3) - H(X_1) - H(X_2). \end{aligned}$$

Step 2: three folds of P_5 onto P_3.

Label the vertices of P_3 by

$$0, 1, 2, 3$$

in their natural order.

Consider the following three graph homomorphisms $\phi_i : P_5 \to P_3$, specified by the images of the six consecutive vertices of P_5:

$$\begin{aligned} \phi_1 &: (0, 1, 0, 1, 2, 3), \\ \phi_2 &: (0, 1, 2, 1, 2, 3), \\ \phi_3 &: (0, 1, 2, 3, 2, 3). \end{aligned}$$

Thus ϕ_i traverses every edge of P_3 once, except for the ith edge, which it traverses three times.

We now explain how to "pull back" the distribution of X along each ϕ_i.

More generally, suppose

$$a_0, a_1, \ldots, a_5 \in \{0, 1, 2, 3\}$$

form a walk in P_3. Define a random walk

$$Y = (Y_0, \ldots, Y_5)$$

in G as follows. First sample Y_0 with the same distribution as X_{a_0}. Having sampled Y_j, sample Y_{j+1} according to the conditional distribution

$$\mathbb{P}(Y_{j+1} = v \mid Y_j = u) := \mathbb{P}(X_{a_{j+1}} = v \mid X_{a_j} = u).$$

Then

$$Y_j \overset{d}{=} X_{a_j}$$

for every j, and

$$(Y_j, Y_{j+1}) \overset{d}{=} (X_{a_j}, X_{a_{j+1}})$$

for every j. Moreover, Y is supported on $\mathrm{Hom}(P_5, G)$.

Applying this construction to the three walks ϕ_1, ϕ_2, ϕ_3 gives random homomorphisms

$$Y^{(1)}, Y^{(2)}, Y^{(3)} \in \mathrm{Hom}(P_5, G).$$

We take these three random homomorphisms independently.

Finally, let Z_0 and Z_3 be independent random vertices having, respectively, the distributions of X_0 and X_3, and independent of all the $Y^{(i)}$.

Altogether,

$$Y := (Z_0, Z_3, Y^{(1)}, Y^{(2)}, Y^{(3)})$$

is a random element of $\mathrm{Hom}(F, G)$.

Step 3: compute the entropy.

For any Markov distribution on a path $v_0 v_1 \cdots v_m$, one has

$$H(V_0, \ldots, V_m) = \sum_{j=0}^{m-1} H(V_j, V_{j+1}) - \sum_{j=1}^{m-1} H(V_j). \tag{2}$$

This follows immediately from the chain rule:

$$\begin{aligned} H(V_0, \ldots, V_m) &= H(V_0) + \sum_{j=0}^{m-1} H(V_{j+1} \mid V_j) \\ &= H(V_0) + \sum_{j=0}^{m-1} \big(H(V_j, V_{j+1}) - H(V_j)\big). \end{aligned}$$

We apply (2) to the three copies of P_5.

Across the three maps ϕ_1, ϕ_2, ϕ_3, each edge

$$01, \qquad 12, \qquad 23$$

of P_3 occurs exactly five times: each occurs once in each ϕ_i, and twice more in the unique ϕ_i which backtracks over that edge.

Hence the positive edge contribution to

$$H(Y^{(1)}) + H(Y^{(2)}) + H(Y^{(3)})$$

is

$$5H(X_0, X_1) + 5H(X_1, X_2) + 5H(X_2, X_3).$$

It remains to count the internal vertices of the three copies of P_5. Their images under the three ϕ_i are

ϕ_1	$1, 0, 1, 2$
ϕ_2	$1, 2, 1, 2$
ϕ_3	$1, 2, 3, 2.$

Thus, among all internal vertices,

$$0 \text{ occurs once}, \qquad 1 \text{ occurs five times}, \qquad 2 \text{ occurs five times}, \qquad 3 \text{ occurs once}.$$

Therefore

$$\sum_{i=1}^{3} H(Y^{(i)}) = 5H(X_0, X_1) + 5H(X_1, X_2) + 5H(X_2, X_3) \\ - H(X_0) - 5H(X_1) - 5H(X_2) - H(X_3).$$

The two isolated vertices contribute

$$H(Z_0) + H(Z_3) = H(X_0) + H(X_3).$$

Since all five components were chosen independently,

$$\begin{aligned} H(Y) &= H(Z_0) + H(Z_3) + \sum_{i=1}^{3} H(Y^{(i)}) \\ &= 5\Big(H(X_0, X_1) + H(X_1, X_2) + H(X_2, X_3) - H(X_1) - H(X_2)\Big). \end{aligned}$$

By (1),

$$H(Y) = 5H(X).$$

Since X is uniform on $\mathrm{Hom}(P_3, G)$,

$$H(Y) = 5 \log \hom(P_3, G).$$

On the other hand, Y is supported on $\mathrm{Hom}(F, G)$, so the maximum-entropy bound gives

$$\begin{aligned} 5 \log \hom(P_3, G) &= H(Y) \\ &\le \log |\mathrm{Hom}(F, G)| \\ &= \log\big(n^2 \hom(P_5, G)^3\big). \end{aligned}$$

Exponentiating,

$$\hom(P_3, G)^5 \le n^2 \hom(P_5, G)^3.$$

Dividing by n^{20} yields

$$t(P_3, G)^5 \le t(P_5, G)^3.$$

Finally, the inequality extends from finite graphs to graphons by the standard approximation of graphons by finite graphs together with continuity of homomorphism densities. Hence, for every graphon W,

$$\boxed{t(P_5, W)^3 \ge t(P_3, W)^5}.$$

□

2 The same argument on graphons

The proof of Section 1 ends by transferring the finite inequality to graphons "by the standard approximation of graphons by finite graphs together with continuity of homomorphism densities". That step is delicate at the generality the catalogue needs: its graphons live on an arbitrary probability space, not necessarily atomless or standard Borel, and the usual step-function approximation is not available there. The honest finite-to-graphon route is the W-random sampling argument

of `notes/triangle_three_edge_tail.pdf`, which needs a product space, a conditional Bernoulli construction, a second-moment estimate and convergence in probability.

None of that is necessary. The entropy proof transcribes directly. What disappears is exactly the appeal to a uniform distribution on a finite set: Shannon entropy is replaced by relative entropy with respect to μ, the maximum-entropy bound $H(Y) \le \log|\mathrm{Hom}(F,G)|$ becomes Gibbs' inequality, and the two isolated vertices — present in Section 1 only to supply the factor n^2 — become two applications of the nonnegativity of relative entropy.

2.1 Setting

Let (Ω,μ) be an arbitrary probability space and $W\colon \Omega^2 \to [0,1]$ a symmetric measurable graphon. Write $T_W h(x) = \int W(x,y)h(y)\,\mathrm{d}\mu(y)$ and

$$d = T_W\mathbf{1}, \qquad A = T_W^2\mathbf{1}, \qquad B = T_W^3\mathbf{1},$$

$$a_3 = t(P_3,W) = \langle \mathbf{1}, T_W^3\mathbf{1}\rangle, \qquad a_5 = t(P_5,W) = \langle \mathbf{1}, T_W^5\mathbf{1}\rangle.$$

Assume until Section 2.6 that $W \ge \varepsilon$ pointwise for some $\varepsilon > 0$; then $d, A, B \ge \varepsilon$, every density below is bounded above and below by positive constants, and every logarithm appearing is bounded. All integrability hypotheses are therefore automatic, and we do not comment on them again.

2.2 The P_3 measure and its marginals

Let

$$f(x_0,x_1,x_2,x_3) = \frac{W(x_0,x_1)W(x_1,x_2)W(x_2,x_3)}{a_3},$$

a probability density on Ω^4 with respect to μ^4. This is the graphon counterpart of "a uniformly random homomorphism $P_3 \to G$". Integrating out coordinates gives its one- and two-dimensional marginal densities in closed form:

$$\begin{gathered} m_0 = m_3 = \frac{B}{a_3}, \qquad m_1 = m_2 = \frac{d\,A}{a_3}, \\ M_{01}(u,v) = \frac{W(u,v)A(v)}{a_3}, \qquad M_{12}(u,v) = \frac{W(u,v)d(u)d(v)}{a_3}, \qquad M_{23}(u,v) = \frac{W(u,v)A(u)}{a_3}. \end{gathered} \tag{3}$$

Each is a probability density: for instance $\iint M_{01} = a_3^{-1}\int d(v)A(v)\,\mathrm{d}\mu(v) = a_3^{-1}\langle T_W\mathbf{1}, T_W^2\mathbf{1}\rangle = 1$. For a reversed orientation write $M_{10}(u,v) = M_{01}(v,u)$, and similarly for the other edges.

The Markov property of Step 1 of Section 1 becomes the exact factorization

$$f = \frac{M_{01}M_{12}M_{23}}{m_1\,m_2}, \tag{4}$$

which one checks by substituting (3) and cancelling $d(x_1)A(x_1)d(x_2)A(x_2)$.

2.3 The one analytic input

Lemma 2.1 (Gibbs)**.** *Let ν be a measure and let $f, g > 0$ be probability densities with respect to ν. Then $\int f\log(g/f)\,\mathrm{d}\nu \le 0$.*

Proof. Pointwise $\log u \le u - 1$, so $f\log(g/f) \le f\,(g/f - 1) = g - f$, and $\int (g-f)\,\mathrm{d}\nu = 0$. □

Taking $g \equiv 1$ gives the only other fact we need:

$$\int m \log m \, \mathrm{d}\nu \;\geq\; 0 \qquad \text{for every probability density } m. \tag{5}$$

This is what replaces "$H(Z_0) \leq \log n$" in Section 1.

2.4 The three folds

Let $a = (a_0, \ldots, a_5)$ be a walk in P_3 and put

$$q_a(y_0, \ldots, y_5) = \frac{\prod_{j=0}^{4} M_{a_j a_{j+1}}(y_j, y_{j+1})}{\prod_{j=1}^{4} m_{a_j}(y_j)}. \tag{6}$$

Lemma 2.2. *q_a is a probability density on Ω^6; its marginal on the coordinate pair $(j, j+1)$ is $M_{a_j a_{j+1}}$, and its marginal on the coordinate j is m_{a_j}.*

Proof. Integrate out y_5 first: $\int M_{a_4 a_5}(y_4, y_5)\, \mathrm{d}\mu(y_5) = m_{a_4}(y_4)$, which cancels the factor $m_{a_4}(y_4)$ in the denominator and leaves the same expression for the walk $(a_0, \ldots, a_4)$. Iterating down to a single coordinate leaves m_{a_0}, of total mass one. The same computation performed from either end gives the stated marginals. □

This is (6) written out; no conditional-probability machinery is used, which is what makes the argument valid on an arbitrary probability space. Set

$$E_e = \iint M_e \left(\log W - \log M_e\right) \mathrm{d}\mu^2 \quad (e \in \{01, 12, 23\}), \qquad V_k = \int m_k \log m_k \, \mathrm{d}\mu.$$

Note E_e does not depend on the orientation of e, because W is symmetric.

Lemma 2.3 (Tree-entropy identity)**.** $E_{01} + E_{12} + E_{23} + V_1 + V_2 = \log a_3$.

Proof. Compute $\int f \log f$ in two ways. From $f = W_{01} W_{12} W_{23} / a_3$ and Lemma 2.2 applied to the walk $(0, 1, 2, 3)$,

$$\int f \log f = \sum_e \iint M_e \log W - \log a_3.$$

From the factorization (4),

$$\int f \log f = \sum_e \iint M_e \log M_e - V_1 - V_2.$$

Subtract. □

Lemma 2.4 (Per-fold bound)**.** *For every walk a in P_3 of length five,*

$$\log a_5 \;\geq\; \sum_{j=0}^{4} E_{\{a_j, a_{j+1}\}} + \sum_{j=1}^{4} V_{a_j}.$$

Proof. Apply Lemma 2.1 on Ω^6 with $f = q_a$ and $g = a_5^{-1} \prod_{j=0}^{4} W(y_j, y_{j+1})$, both probability densities. Expanding $\log(g/q_a)$ by (6) and integrating term by term with Lemma 2.2,

$$0 \;\geq\; \int q_a \log \frac{g}{q_a} = \sum_{j=0}^{4} \iint M_{a_j a_{j+1}} \left(\log W - \log M_{a_j a_{j+1}}\right) + \sum_{j=1}^{4} \int m_{a_j} \log m_{a_j} - \log a_5. \qquad \square$$

2.5 Summation

Take the three folds of Section 1,

$$\phi_1 = (0,1,0,1,2,3), \qquad \phi_2 = (0,1,2,1,2,3), \qquad \phi_3 = (0,1,2,3,2,3),$$

and add the three instances of Lemma 2.4. The multiplicities are those already counted there: each edge of P_3 is traversed $3+1+1=5$ times in total, and among the twelve internal vertices the labels $0,1,2,3$ occur $1,5,5,1$ times. Hence

$$3\log a_5 \;\geq\; 5\big(E_{01}+E_{12}+E_{23}\big) + V_0 + 5V_1 + 5V_2 + V_3 \;=\; 5\log a_3 + V_0 + V_3$$

by Lemma 2.3, and $V_0, V_3 \geq 0$ by (5). Therefore

$$a_5^3 \;\geq\; a_3^5 \qquad \text{whenever } W \geq \varepsilon > 0. \tag{7}$$

2.6 Removing the regularization

For a general graphon W and $0 < \varepsilon < 1$ put $W_\varepsilon = (1-\varepsilon)W + \varepsilon$, again a graphon, with $W_\varepsilon \geq \varepsilon$. Since $\|W_\varepsilon - W\|_\infty \leq 2\varepsilon$ and a homomorphism density with m edges is m-Lipschitz in the supremum norm,

$$|t(P_k, W_\varepsilon) - t(P_k, W)| \leq 2k\varepsilon.$$

Applying (7) to W_ε and letting $\varepsilon \downarrow 0$ gives $t(P_5,W)^3 \geq t(P_3,W)^5$ for every graphon W on every probability space. □

Remark 2.5. *The two isolated vertices of $F = P_0 \sqcup P_0 \sqcup P_5 \sqcup P_5 \sqcup P_5$ do not appear above. In Section 1 they contribute the factor n^2 that compensates $-H(X_0) - H(X_3)$; here μ is already a probability measure, $t(P_0, W) = 1$, and the same compensation is supplied by $V_0, V_3 \geq 0$. This is the only structural difference between the two proofs.*

Remark 2.6. *Every step above is finite-dimensional integration against explicit densities built from W, d, A, B and a_3, together with one scalar inequality (Lemma 2.1) and one limit. In the Lean development Lemma 2.1 is `integral_mul_log_div_nonpos` of `Methods/PureChordal/Gibbs.lean`, the limit of Section 2.6 uses `Methods/PureChordal/Regularization.lean`, and the coordinate integrations of Lemma 2.2 use the assignment-integral peeling ladder. The same combination already carries the entropy proof of the pure-chordal theorem in `Methods/PureChordal/EntropyGluing.lean`.*

A What the Lean formalization actually proves

Section 2 is formalized in full, as `a3_pow_five_le_a5_pow_three` in `lean/Taeyoung/Methods/OddWalk/`, axiom-clean. One difference of substance.

No measure on Ω^4 or Ω^6 is constructed

The argument above is written against the probability density (4) on Ω^4, and Lemma 2.2 integrates a six-coordinate quantity. Neither space is built. Everything is phrased instead inside Ω, through the iterate $T_W^n \mathbf{1}$, with $a_3 = \int B$ and $a_5 = \int AB$ by self-adjointness of T_W. Three consequences:

- The marginals (3) are supplied as four closed-form functions — P_3 has two kinds of vertex and two kinds of edge — rather than as marginals of anything. The remaining two are these with their arguments interchanged.

- The step inequality of Lemma 2.4 is applied *pointwise in the outer variable*: for fixed y the conditional density $z \mapsto K(y,z)/u(y)$ is a probability density, which is the Markov property of the folded walk, so each step is one application of Lemma 2.1 on Ω and one Fubini on Ω^2. This is what replaces the six-fold construction that Section 1 performs when it samples a random homomorphism.

- The tree-entropy identity of Lemma 2.3 is not obtained by evaluating $\int f \log f$ in two ways. Because the marginals are explicit quotients, every logarithm splits and the identity becomes scalar algebra in
$$P_d = \int_\Omega \frac{dA}{a_3} \log d, \qquad P_A = \int_\Omega \frac{dA}{a_3} \log A :$$
the $\log W$ cancels inside each edge term, leaving $E_{01} = E_{23} = \log a_3 - P_A$, $E_{12} = \log a_3 - 2P_d$ and $V_1 = V_2 = P_d + P_A - \log a_3$, whose combination is $\log a_3$.

Accordingly the closing remark of Section 2.6 is corrected on one point: the coordinate integrations of Lemma 2.2 do *not* use the assignment-integral peeling ladder, which the development never invokes.